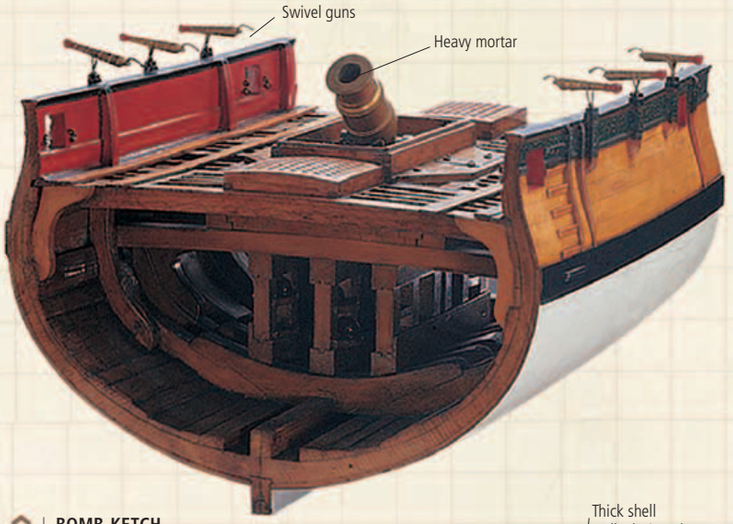
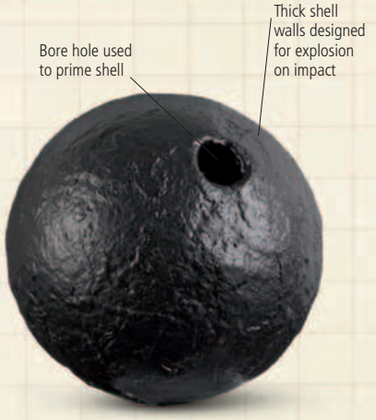
SHOT TO SHELL

UNTIL THE 19TH CENTURY naval cannon fired varieties of solid shot. Explosive shells—hot shot filled with gunpowder—were employed in land warfare, but they were thought unsafe for use at sea. From the 1830s, navies adopted shells as part of their munitions. The introduction of iron armor on warships around 1850 made solid shot obselete—it simply bounced off the cladding. By the 1880s rifled breech loading guns firing high-exposive shells had become standard.



BOMB-KETCH
These were developed to carry heavy naval mortars. Here the mortar projects through a hatch in the deck to deliver its bombardment.



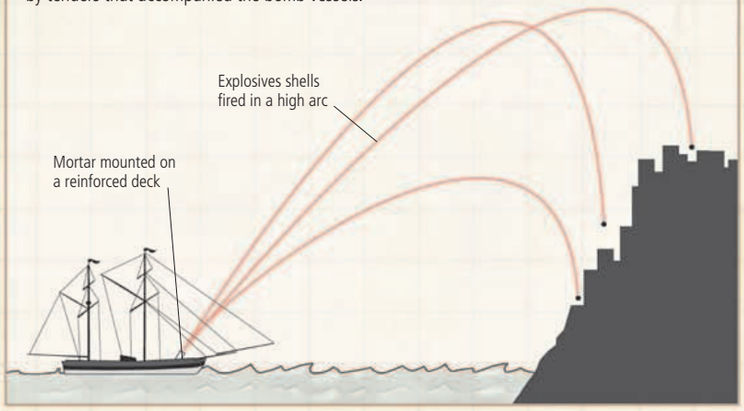
EXPLOSIVE SHELL
A bore hole was used to fill the shell with a small charge of gunpowder, which would then be detonated via a time-delay fuse.



SHELL CUTAWAY
Firing the shell ignited a slow-burning fuse set into the bore hole, which then ignited the gunpowder. A wooden plug, or sabot, kept the fuse centered in the barrel during shell loading.

OFFSHORE BOMBARDMENT

Bombarding targets on shore has always been a major application of ships' guns. In land warfare, mortars and howitzers were developed for the bombardment of besieged towns and fortresses. These were guns firing in a high arc, using "bombs" or explosive shells as their munitions. The French navy was the first to develop vessels designed to mount mortars in the 1680s, and was soon imitated by other navies. These bomb galiots or bomb ketches were built with reinforced structures to cope with the forces released by the guns. To avoid the danger of keeping stores of explosive shells on a vessel that had guns firing, the shells were usually carried by tenders that accompanied the bomb vessels.

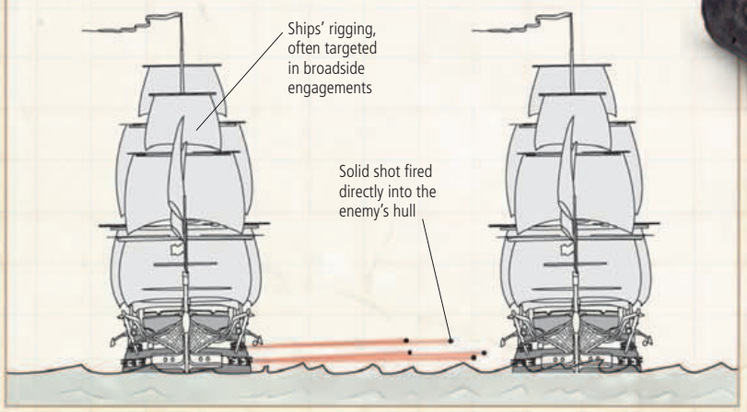




SOLID SHOT
Before shells, shot was mainly used for penetrating heavy structures, in set-piece engagements. It had a formidable effect on personnel, guns, and equipment.

SHIP VERSUS SHIP GUN DUELS

In the age of sail, guns on warships had a limited range. While an 18th century cannon could fire traditional solid shot up to 1 mile (1.6 km), its flat trajectory (the most accurate part of its flight) was about 656 ft (200 m). Just to be sure of the shot, however, vessels would frequently close to half that distance before opening up with huge broadsides. Different navies employed different tactics in gun engagements. The Royal Navy, for example, preferred to shoot into the enemy's hull to kill men or sink the craft, while the French Navy frequently concentrated on masts, sails, and rigging in an attempt to hinder a ship's maneuverability.





CANISTER
The metal canister here would rupture as it left the muzzle of the cannon, allowing the small shot inside to continue in a wide formation against enemy personnel.